Sura Rohith Sai

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.67
Narayana Junior College	Class XII (Telangana State Board)	2024	94.00%
Takshasila Public School	Class X (CBSE)	2022	95.20%

SKILLS

Programming Languages: C, C++, Python, Verilog, MATLAB

EDA Tools: Altium Designer, LTspice, Icarus Verilog, GTKWave

Development Tools: Git, GitHub, VS Code, Arduino IDE, LaTeX

Operating Systems: Windows, Mac

PROJECTS

RISC-V RV32I Processor -- RTL Design & Verification | github.com/Rohith286/RISC-V

- Designed and implemented a 32-bit single-cycle RISC-V (RV32I) processor in Verilog, including the Program Counter, Register File, ALU, Immediate Generator, Control Unit, and Data Path
- Extended the design into a 5-stage pipelined RV32I processor (IF, ID, EX, MEM, WB) with pipeline registers and staged control/data signal propagation
- Resolved data hazards using forwarding paths and hazard detection logic, including stall insertion for load-use hazards
- Verified individual RTL modules with self-checking testbenches, then integrated and validated the pipeline against the RV32I base instruction set

Parameterizable UART Transceiver -- RTL Design & Verification | github.com/Rohith286/UART

- Implemented a parameterizable UART receiver in Verilog using a 4-state FSM, with configurable system clock frequency, baud rate, and payload width.
- Implemented start-bit glitch validation, mid-bit data sampling at 7×CLKS_PER_BIT/16, configurable data-bit indexing, and stop-bit validation with frame-error detection.
- Verified UART functionality using self-checking loopback testbenches with pass/fail checks and timeout guards, and debugged simulation waveforms using Icarus Verilog and GTKWave.

8-Point FFT Processor -- Verilog RTL

- Implemented an 8-point radix-2 DIT FFT processor in Verilog as a 3-stage pipelined butterfly network with streaming I/O, using Q3.5 fixed-point twiddle factor coefficients
- Verified functional correctness against MATLAB-generated reference outputs, achieving a maximum error of 0.03 consistent with the fixed-point quantization step
- Synthesized the design and benchmarked against Intel's FFT IP core, achieving comparable per-clock throughput with significantly lower resource utilization (logic, DSP blocks, memory)

Pipelined FIR Filter -- Verilog RTL

- Designed a parameterized pipelined FIR filter in Verilog using Q2.14 fixed-point arithmetic, achieving single-cycle throughput via a pipelined adder tree.
- Verified filter frequency response and numerical accuracy against MATLAB reference using fixed-point test vectors.

8-Bit Computer -- Breadboard Implementation

- Built a fully functional 8-bit computer on a breadboard using discrete logic ICs, implementing a custom ALU, registers, program counter, memory interface, and hardwired control unit.

- Debugged the hardware at the gate level and executed instruction sets via manually programmed microcode sequences.

RELEVANT COURSES

Core Courses: Analog Electronics, Digital Systems, Circuit and Network Analysis, Signals & Systems, Digital Signal Processing, Semiconductor Device Fundamentals, Control Systems

Computer Science Courses: Introduction to Programming, Data Structures & Algorithms, Database Management Systems, Machine Learning.

Mathematics Courses: Probability & Random Processes, Matrix Theory, Differential Equations & Transform Techniques.

Laboratory Courses: Digital Systems Laboratory, Electronic Devices & Circuits Laboratory, Digital Signal Processing Laboratory, PCB Design & Layout for Electronic Systems.

POSITIONS OF RESPONSIBILITY

Production & Pronites Coordinator, Elan & nVision, IIT Hyderabad

- Coordinated production planning and on-ground execution for large-scale technical and cultural events.
- Managed logistics, scheduling, and cross-functional team coordination to ensure seamless event operations.

Hospitality & Logistics Coordinator, SPIC MACAY, IIT Hyderabad

- Coordinated hospitality, accommodation, and logistics for artists and guests during cultural events.
- Collaborated with organizing teams to ensure smooth event execution and participant experience.

EXTRACURRICULAR

Basketball - Represented the institute in inter-college basketball tournaments and participated in multiple intra-college competitions.